Kai Liu

Publication List

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Published Articles

1. Phase retrievability of frames and quantum channels, **Kai Liu** and Deguang Han, *Linear Algebra and its Applications*, Vol. 725, pp. 378–400, 2025. PDF
2. Frame dimension functions and phase retrievability, Deguang Han and **Kai Liu**, *Advances in Operator Theory*, Vol. 9, Article 31, 2024. PDF
3. Zero-error correctibility and phase retrievability for twirling channels, Deguang Han and **Kai Liu**, *Reports on Mathematical Physics*, Vol. 93, No. 1, pp. 87–102, 2024. PDF
4. Orbit-injective covariant quantum channels, **Kai Liu**, Chuangxun Cheng, and Deguang Han, *Linear Algebra and its Applications*, Vol. 668, pp. 28–50, 2023. PDF

Accepted for Publication

5. Enhancing Phase Retrievability of Quantum Channels via Interferometric Coupling, **Kai Liu**, Deguang Han, and Omar Nour, arXiv:2604.24363, 2026. PDF
   Accepted by *Journal of Physics A: Mathematical and Theoretical* on October 12, 2026. Online publication forthcoming.

Revised Manuscripts Under Review

6. Fidelity preserving and decoherence for mixed unitary quantum channels, **Kai Liu** and Deguang Han, arXiv:2410.20117, 2024; revised 2026. Revision submitted; **under review** by *Quantum Information Processing*. PDF

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7. Intermediate memory costs of quantum channel synthesis with a protected subsystem, **Kai Liu** and Deguang Han, arXiv:2610.20536, 2026.
   Under review by*Physical Review A*.PDF
8. Vanishing Moment Dilations and Applications to Balanced and Graph Frames, Deguang Han, Shankhadeep Mondal, **Kai Liu**, and Omar Nour, arXiv:2610.11043, 2026. **Under review** by *Journal of Fourier Analysis and Applications*. PDF
9. Extreme Points of Optimal Dual Frames for Erasures, Shankhadeep Mondal, Deguang Han, and **Kai Liu**. **Under review** by *Sampling Theory, Signal Processing, and Data Analysis*.
10. Frame phase retrievability and state distinguishability of quantum channels, Deguang Han and **Kai Liu**, arXiv:2608.30006, 2026. **Under review** as a book chapter. PDF

Ph.D. Thesis

11. Phase Retrieval and Fidelity Preserving Quantum Channels, **Kai Liu**, Ph.D. dissertation, University of Central Florida, 2025. PDF